

NAVEGANTES, Brazil

WMO: 825991

Lat: 26.880S

Lon: 48.651W

Elev: 6

StdP: 101.26

Time Zone: -3.00 (SAB)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	10.3	5.0	5.4	12.0	6.9	6.2	13.5	8.5	16.9	7.6	16.4	4.4	260	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	32.0	26.8	30.9	26.6	29.9	26.2	28.1	30.1	27.3	29.4	26.7	28.8	5.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.8	23.8	29.5	26.9	22.6	28.8	26.1	21.5	28.2	90.2	29.6	86.6	29.8	84.3	29.1	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.3	6.6	5.7	36.1	1.5	1.4	4.6	37.1	3.7	37.9	2.9	38.7	1.8	39.6		
(5)			5.3	29.2	1.6	1.1	4.1	29.9	3.2	30.6	2.3	31.2	1.1	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	26.2	26.2	24.9	23.2	20.5	18.2	17.4	18.0	19.5	21.3	22.8	24.7		
	DBStd	3.78	1.77	1.75	1.75	2.43	2.27	2.70	2.70	2.69	2.19	1.94	1.78	2.14		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	145	0	0	0	2	7	35	48	39	13	1	0	0		
	CDD10.0	4341	503	453	463	396	327	246	229	249	286	351	385	456		
	CDD18.3	1446	244	220	204	148	75	31	19	29	49	94	135	197		
	CDH23.3	8811	2201	1954	1478	808	169	51	23	46	53	182	511	1334		
	CDH26.7	1922	639	520	286	112	14	6	3	8	3	17	36	278		
(14) Wind	WSAvg	3.5	3.9	3.6	3.6	3.2	3.1	3.0	2.9	3.2	3.4	3.6	4.1	3.9		
(15) Precipitation	PrecAvg	1779	260	200	183	94	106	109	128	89	148	171	136	158		
	PrecMax	2523	392	317	292	210	364	217	307	291	310	348	255	245		
	PrecMin	1440	149	70	57	36	15	54	20	13	17	73	74	47		
	PrecStd	264	72	72	59	47	94	36	73	75	71	66	51	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	33.0	31.9	31.0	28.1	27.1	26.1	27.8	26.9	28.9	30.0	33.2		
		MCWB	27.0	27.1	26.0	26.2	24.5	21.4	21.8	22.1	22.1	23.1	23.4	26.1		
	2%	DB	32.0	31.2	30.0	29.1	26.1	24.8	23.1	24.1	24.9	26.2	27.2	30.8		
		MCWB	27.1	27.4	26.0	26.0	22.8	20.7	19.5	20.7	21.9	22.9	23.8	26.3		
	5%	DB	30.9	30.2	29.0	27.9	25.0	23.1	22.0	22.9	23.8	25.1	26.2	29.1		
		MCWB	26.8	26.8	25.7	24.9	22.2	20.2	19.4	20.7	21.2	21.9	23.2	25.5		
	10%	DB	29.9	29.2	28.1	27.0	24.0	21.9	21.0	21.2	22.8	24.1	25.9	28.1		
		MCWB	26.1	26.2	24.8	24.2	21.7	19.6	19.0	19.3	20.6	21.3	23.0	24.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.8	28.9	28.1	29.1	25.3	23.1	22.2	23.2	24.0	26.1	27.0	28.1		
		MCDB	31.5	31.1	29.3	29.9	27.3	25.6	24.6	24.4	25.1	26.2	27.0	30.9		
	2%	WB	27.8	28.0	26.9	27.1	23.9	21.9	20.6	21.9	22.5	24.0	25.3	26.9		
		MCDB	30.6	30.0	28.7	27.9	25.3	23.6	22.1	23.8	23.9	25.1	26.0	29.2		
	5%	WB	27.0	27.3	26.2	26.0	23.0	20.9	19.8	20.9	21.7	22.9	24.4	26.2		
		MCDB	29.8	29.5	28.0	26.8	24.4	22.4	21.3	22.2	23.0	24.1	25.5	28.3		
	10%	WB	26.3	26.6	25.6	24.9	22.2	20.2	19.3	20.1	21.0	22.1	23.6	25.4		
		MCDB	28.9	28.9	27.5	26.1	23.6	21.5	20.8	21.4	22.2	23.6	25.1	27.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.1	6.3	6.7	6.4	6.9	6.7	6.6	5.6	5.0	5.5	5.8		
		MCDBR	8.4	7.2	7.1	7.2	7.4	8.3	7.8	7.8	6.5	6.2	6.2	7.8		
	5% WB	MCWBR	4.6	4.1	4.1	4.3	4.4	4.9	4.8	4.7	4.0	3.8	3.8	4.3		
		MCWBR	7.5	6.6	6.7	6.1	6.5	6.9	6.8	7.0	5.7	5.3	5.6	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.415	0.408	0.397	0.390	0.371	0.390	0.393	0.460	0.523	0.464	0.424	0.419		
		taud	2.439	2.458	2.482	2.482	2.506	2.441	2.415	2.174	1.992	2.229	2.361	2.404		
	Ebn at Noon		929	919	899	858	829	783	796	773	769	858	915	928		
		Edh at Noon	123	117	109	100	89	91	96	133	175	146	132	127		
(44) All-Sky Solar Radiation	RadAvg		5.68	5.40	4.71	3.88	3.01	2.58	2.77	3.32	3.72	4.28	5.22	5.59		
	RadStd		0.42	0.47	0.46	0.37	0.33	0.21	0.33	0.34	0.47	0.61	0.53	0.50		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	-0.28	-0.44	N/A	N/A	N/A	N/A

Nomenclature: See separate page